

5. PEST MANAGEMENT ISSUES IN MOZAMBIQUE

This section describes the current and anticipated pest management problems relevant to the Project, relevant IPM experience within the project area, assessment of proposed or current pest and pesticide management approaches and recommendations. The section considers the fact that the Project will have a strong agricultural development component, particularly under “Agriculture Intensification and Market Linkages”. The latter has been the area in which the GOM’s ability to diversify the economy and deepen inclusion has been significantly weak. The contribution of the agriculture sector to the country’s GDP has been far below its real potential. Strategic and policy documents for the sector adopted lately such as PEDSA (Strategic Plan for Agricultural Development), PNISA (National Agriculture Investment Plan), PNI (National Irrigation Program), aim at reversing this phenomenon and IRRIGA embraces them.

5.1 Current and anticipated pest problems

The use of pesticides is likely to be associated with agricultural activities (plant production) for the control of pests, diseases, nematodes and weeds. Main crops targeted under IPMP in the project area and particularly the irrigated schemes to be established are likely to be **rice, maize, sugar cane**, a variety of **beans, horticultures and fruits**³⁷. Other crops are likely to be adopted as their feasibility becomes apparent. These are associated with several plant diseases typical of the Mozambican context. Different ways of combating such diseases are adopted by the three main categories farmers, namely (i) the family sector; (ii) the small and medium size operators; and (iii) large farmers and operators. The expected intensification involving the two first categories of farmers under IRRIGA is likely to trigger the application of agrochemicals and inputs such as fertilizers, herbicides, insecticides, nematicides and fungicides. This has been happening under PROIRRI and it is expected to continue and if not controlled even to increase.

In the context of this text and in Mozambique in general pests affecting plants are subdivided into three main categories i.e. pests, diseases and infestants. **Error! Reference source not found., Error! Reference source not found.,** and Table 5-3 successively present the most common pests, infestants and diseases and associated crops in the country. Infestants (weeds) tend to be associated with all kinds of crops and are widespread.

Table 5-1: List of major pests and associated crops in Mozambique

Common Name	Name of Pest or Vector	Crop
Insects		
Brocas do colmo	<i>Busseola fusca</i> , <i>Sesamia calamistis</i> (Lepidoptera: Noctuidae)	Maize, Rice & Sugar cane
	<i>Chilo partellus</i> (Lepidoptera: Crambidae)	Maize, Rice & Sugar cane
Brocas da espiga	<i>Mussidia nigrivenella</i> (Lepidoptera: Pyralidae)	Maize
Lagarta do funil do milho Fall armyworm	<i>Spodoptera frugiperda</i> (Lepidoptera: Noctuidae)	Maize

³⁷ Under PROIRRI except for green beans there was not enough consideration for this crop.

Common Name	Name of Pest or Vector	Crop
Lagarta das searras ou militar	<i>Spodoptera exempta</i> = <i>Laphygna exempta</i> (Lepidoptera: Noctuidae)	Maize
Pragas de armazenamento	<i>Sitotroga cerealella</i> (Lepidoptera: Gelechiidae)	Maize
	<i>Sitophilus orizae</i> (Coleoptera: curculionidae)	Maize
Roscas (lagartas)	<i>Agrotis segetum</i> , <i>A. ypsilon</i> (Lepidoptera: Noctuidae)	Maize
Pássaros (pedizes, etc.)	<i>Rhynchotus rufescens</i>	Maize & Beans
Ratos, Toupeiras	<i>Talpidae</i>	Maize & Beans
Ácaros	<i>Tetranychus urticae</i>	Beans, Cowpea & Vegetable
Besouro da flor	<i>Mylabris</i> spp, <i>Coryna</i> spp. (Coleoptera: Meloidae)	Beans & Cowpea
Besouro da folhagem	<i>Ootheca mutabilis</i> , <i>O. bennigseni</i> (Coleoptera: Chrysomelidae)	Beans & Cowpea
Besouro listrado	<i>Alcidodoles leucogrammus</i> (Coleoptera: Curculionidae)	Beans & Cowpea
Cigarrinha verde	<i>Empoasca dolichi</i> , <i>E. lybica</i> (Hemiptera: Cicadellidae)	Beans & Cowpea
Gorgulhos	<i>Acanthoscelides obtectus</i> & <i>Zabrotes subfasciatus</i> (Coleoptera: Bricidae)	Beans & Cowpea
Lagarta caterpillar	<i>Spodoptera</i> spp. (Lepidoptera: Noctuidae)	Beans & Cowpea
Lagarta das cápsulas do algodoeiro	<i>Helicoverpa (Heliothis) armigera</i> (Lepidoptera: Noctuidae)	Beans & Cowpea
Lagarta-das-vagens	<i>Marura testulalis</i> (Lepidoptera: Pyraustidae)	Beans & Cowpea
Lagartas (roscas)	<i>Agrotis segetum</i> & <i>A. ypsilon</i> (Lepidoptera: Aleyrodidae)	Beans & Cowpea
Mosca branca	<i>Bemisia tabaci</i> (Hemiptera: Aleyrodidae)	Beans & Cowpea, vegetable
Mosca do feijoeiro	<i>Ophiomyia phaseoli</i> , <i>O. spencerella</i> (Diptera: Agromyzidae)	Beans & Cowpea
Pulgão preto	<i>Aphis fabae</i> , <i>A. cracivora</i> <i>Myzus persicae</i> (Hemiptera: Aphididae)	Beans, Cowpea & vegetable
Trips	<i>Mgalurothrips sjostedti</i> ; <i>Thrips tabaci</i> (Lindeman)	Beans, Cowpea & vegetable
Diamond back moth	<i>Plutella xylostella</i>	Brassicas
Traca do tomate	<i>Tuta absoluta</i>	Tomato
Mosca das frutas	<i>Bactrocera</i> spp	Tomato & other fruits
Lagarta Americana = Africana	<i>Helicoverpa zea</i> (Lepidoptera: Noctuidae)	Vegetable

Table 5-2: List of major weeds (infestants) in Mozambique

Family name	Scientific name	Classifier
	<i>Pteridophyta</i>	
Acanthaceae	<i>Justiça sp</i>	
	<i>Dicotyledoneae</i>	
Amaranthaceae	<i>Amaranthus hibridus</i>	L.
Asteraceae	<i>Acanthospermum xanthioides</i>	DC.
Asteraceae	<i>Ageratum conyzoides</i>	L.
Asteraceae	<i>Bidens biternata</i>	(Lour.) Merr.&Scherff

Family name	Scientific name	Classifier
Asteraceae	<i>Bidens pilosa</i>	L.
Asteraceae	<i>Bidens steppia</i>	(Steetz) Sherff
Asteraceae	<i>Blumea lacera</i>	(Burm.f) DC
Asteraceae	<i>Crassocephalum rubens</i>	(Juss. Ex Jacq.) S. Moore
Asteraceae	<i>Crassocephalum sarcobasis</i>	(DC) S. Moore
Asteraceae	<i>Conyza stricta</i>	Wild
Asteraceae	<i>Emilia coccinea</i>	(Sims) G. Dan
Asteraceae	<i>Feliaia muricata</i>	Thunb.
Asteraceae	<i>Galinsoga parviflora</i>	Cav.
Asteraceae	<i>Pseudognaphalim luteo-album</i>	(L.)
Asteraceae	<i>Tagete minuta</i>	L.
Asteraceae	<i>Vernonia petersii</i>	Oliv. & Hiern
Asteraceae	<i>Vernonia poskeana</i>	Vatke & Hildebrandt
Capparaceae	<i>Cleome iberidella</i>	Welw.
Capparaceae	<i>Cleome monophylla</i>	L.
Chenopodiaceae	<i>Chenopodium ambrosioides</i>	L.
Convolvulaceae	<i>Ipomea eriocarpa</i>	R. Br.
Curcubitaceae	<i>Zehneria racemosa</i>	Hook.f
Fabaceae	<i>Caesalpina sp.</i>	
Fabaceae	<i>Chamaecrista mimisoides</i>	L.
Fabaceae	<i>Crotolaria aculeata</i>	de Wild
Fabaceae	<i>Crotolaria anthyllopsis</i>	Welw.
Fabaceae	<i>Crotolaria comosa</i>	Baker
Fabaceae	<i>Indigostrum costatum</i>	(Guill. & Perr.)
Fabaceae	<i>Indigofera subulifera</i>	Welw.
Fabaceae	<i>Sesbania pachycarpa</i>	DC.
Fabaceae	<i>Tephrosia melanocalix</i>	Welw.
Fabaceae	<i>Vigna sp.</i>	
Lamiaceae	<i>Leucas martinicensis</i>	(Jacq.) R. Br.
Malvaceae	<i>Hibiscus cannabinus</i>	L.
Malvaceae	<i>Sida cardifolia</i>	L.
Menispermaceae	<i>Cissampelos mucronata</i>	A. Rich
Oxalidaceae	<i>Oxalis semiloba</i>	
Portulacaceae	<i>Portulaca oleraceae</i>	L.
Rubiaceae	<i>Calanda rubricaulis</i>	K. Schum
Rubiaceae	<i>Oldenlandia herbacea</i>	(L.) Roxb.
Rubiaceae	<i>Richardia scabra</i>	L.
Scrophulariaceae	<i>Alectra sessiliflora</i>	(vahl) Kuntze
Solanaceae	<i>Datura stramonium</i>	L.
Solanaceae	<i>Nicandra physoloides</i>	(L.) Gaertn
Tiliaceae	<i>Corchorus tridens</i>	L.
Tiliaceae	<i>Lasiosiphon sp</i>	
Tiliaceae	<i>Triumfetta annua</i>	L.
Verbenaceae	<i>Lipia sp</i>	
Commelinaceae	<i>Commelina benghalensis</i>	L.

Family name	Scientific name	Classifier
Commelinaceae	<i>Commelina purpurea</i>	C.B. Clarke
Cyperaceae	<i>Abilgaardia hispidula</i>	(Vahl) Lye
Cyperaceae	<i>Cyperus distans</i>	L.
Cyperaceae	<i>Cyperus esculentus</i>	L.
Cyperaceae	<i>Cyperus rotundus</i>	L.
Cyperaceae	<i>Killinga Odorata</i>	Vahal
Cyperaceae	<i>Mariscus alternifolius</i>	Vahal
Cyperaceae	<i>Mariscus umbellatus</i>	Vahal
Cyperaceae	<i>Mariscus cylindristachyus</i>	Steud
Poaceae	<i>Cynodon dactylon</i>	(L.) Pers.
Poaceae	<i>Digitaria spp</i>	
Poaceae	<i>Eleisine indica</i>	Gaertum
Poaceae	<i>Eragrotis annualata</i>	Rendle
Poaceae	<i>Eragrotis chapelieri</i>	Ness
Poaceae	<i>Eragrotis superba</i>	Peyr
Poaceae	<i>Hyparrhenia rufa</i>	(Nees) Stapf
Poaceae	<i>Melinis repens</i>	(Willd.) Zizka
Poaceae	<i>Setaria sphacelata</i>	(Shumach) Moss
Poaceae	<i>Sporobolus pyramidalis</i>	Beuv

Table 5-3: List of major diseases and pathogenic agents and associated crops in Mozambique

Crop	Disease	Pathogenic Agent
Tomato, Aubergine and Pepper	Seedling wilting	<i>Pythium, Rhizoctonia, Sclerotium e Fusarium</i>
	Concentric spot	<i>Alternaria solani</i>
	Mildew	<i>Phytophthora infestans</i>
	Bacterial wilt	<i>Pseudomonas solanacearum</i>
	Bacterial wilt	<i>Xanthomonas vesicatoria</i>
	Canker	<i>Colletotrichum</i> sp.
	Mosaic and Leaf Winding	Vários vírus
Cabbage, kale/white cabbage, Turnip and Chinese Cabbage	Seedling Wilt	<i>Pythium, Rhizoctonia, Sclerotium e Fusarium</i>
	Mildew	<i>Peronospora parasitica</i>
	Black-rot	<i>Xanthomonas campestris</i>
	Motile Rot	<i>Erwinia carotovora</i>
Reindeer potato	Mildew	<i>Phytophthora infestans</i>
	Concentric spot	<i>Alternaria solani</i>
	Common scab	<i>Streptomyces scabies</i>
	Bacterial Wilt	<i>Pseudomonas solanacearum</i>
	Rhizoctonia (Cancer)	<i>Rhizoctonia solani</i>
	Leaf roll	Potato Leafroll Virus
Onion and garlic	Seedling wilting	<i>Pythium, Rhizoctonia, Sclerotium e Fusarium</i>
	Purple spot	<i>Alternaria porri</i>
	Mildew	<i>Peronospora destructor</i>
	Rust (garlic)	<i>Puccinia allii</i>
	Soft rot (onion)	<i>Botrytis</i> spp.
	White rot	<i>Sclerotium cepivorum</i>
Cucurbitaceas (melon; cucumber; watermelon; pumpkin - Alliaceae: onion and leek).	Seedling wilting	<i>Pythium, Rhizoctonia, Sclerotium e Fusarium</i>
	Mildew	<i>Pseudoperonospora cubensis</i>
	Oidium	Virus (Maize Streak Virus)
	Mosaic	<i>Fusarium; Diplodia</i>

Crop	Disease	Pathogenic Agent
Maize	Mildew	<i>Peronoclenospora</i> sp
	Striped foil	Virus (Maize Streak Virus)
	Cob rot	<i>Fusarium; Diplodia</i>
	Brown stein	<i>Helminthosporium</i> sp
Rice	Burnt	<i>Pyricularia oryzae</i>)
	Brown stein	<i>Drechslera Helminthosporium oryzae</i>
Beans	Mancha angula	<i>Phaeoisariopsis griseola</i>
	Rust	<i>Uromyces appendiculatus</i>
	Foot and rot	<i>Rhizoctonia solani, Sclerotium rolfsii</i>
	Canker	<i>Colletotrichum</i> sp.
	Common mosaic	<i>Bean Common Mosaic Virus</i>
Sugar caner	Mildew	<i>Peronoclenospora</i> sp
	Striped foil	Virus (Maize Streak Virus)
	Brown stain	<i>Helminthosporium</i> sp

In general pest attack is low in the Project targeted areas, but there is a range of pests, diseases and weeds reported by farmers, officials and in the literature. The main factors informing mortality of these pests are (i) climate (cold and dry seasons tends to be better than the hot and wet); (ii) morphology (areas in higher altitude tend to enjoy colder temperatures and increased natural regulation of pests than the lower aeras). These factors work as an opportunity for farmers in general and in irrigated areas. It also follows that the higher areas of Nampula, Zambezia and Manica, which grow horticultures, maize, beans and fruits have more opportunities than the lower areas of Zambezia and Sofala that focus on rice and sugar cane. Farmers already know how to play with these factors and the project will reinforce existing strategies.

The current impact from these pests is not well unknown and/or mapped, except perhaps the red locust that attacks some of the areas. However, the expected pest incidence increases in the production of (i) agriculture-based (rice, maize, sugar cane); (ii) horticulture/fruits will likely reverse that situation and some pests may become a major economic and environmental problem, especially for medium-scale subprojects and, thus an increase in pest control measures and rational use of chemical fertilizers may be needed for this project. There are control measures (chemical, cultural and biological) for most crops that can be used in case an outbreak is observed.

Although the tendency has been to increase, in most areas current pest occurrence and pesticide use is estimated by agricultural officers to be currently low, but an increase in crop area, especially of monoculture crops (rice, sugar cane, fruits and horticulture), may result in an increase of pest occurrence, especially birds (for cereal production areas granivorous birds are an important pest), red locust and rats in the lower areas of Zambezia and Sofala in regard to the latter. These are currently reported to be the major pests in these areas. However, amongst most poor farmers, unless encouraged to do so by different forms of external forces, especially if private operators are given space to do so, pesticide use can be kept at a minimum level if the necessary efforts can be timely made. The prevailing lack of tradition of using pesticides, inadequate supply (the network of agricultural input suppliers, including pesticides is still relatively poor) and poor access to low priced and generic pesticides can be used to work in favor of a strategy to control the massive and indiscriminate use of agro-chemicals.

The development of agricultural activities by a series of farmers, including smallholder farmers and small and medium commercial producers has caused (under PROIRRI) and is likely to cause the intensification of the following potential impacts:

- Stalk borers, brown plant hopper and armyworm could increase, but the result should not be an automatic increase in insecticide application than currently exists in the command area since less than half the farmers apply only one or more sprays per season;
- Irrigated agricultural production is not widespread in the project area now. But with increased irrigation there is the possibility that irrigated cereals (e.g. rice and baby corn) may become important and farmers will be more willing to apply a fungicide to protect their investment.
- Local informants believe fungicide use is unlikely to be greater than presently very low levels that already exists in the area.
- The same consideration applies to weeds, diseases, as well as chemical fertilizer use.

5.2 Relevant IPM experience within the project area and POIRRI

In the same way as in other parts of Mozambique traditional farmers have their own knowledge regarding the use of IPM principles in the Project command area. Different forms of intercropping and/or the use of remedies made from local plants are often adopted with the sole propose of keeping pest at bay and/or increasing soil fertility.

But like in many other areas of traditional knowledge in Mozambique, the mix of it with modern information and practices and poor encouragement for people to use and expand such knowledge explain that it is slowly disappearing. On farm and on station research and extension should endeavor to promote a better understanding of such traditional knowledge and promote what works.

Now and then there is the belief that chemicals are more efficient than any other cultural practices. Though, farmers in several districts prefer crop rotation (summer (e.g. maize) and winter crops (horticulture and a few beans) and intercropping (cereals and vegetables and legumes)) as being efficient in controlling some pests, especially insects and fungus. It is to be expected that these techniques could also be applied in larger areas to be developed under the Project, as they are common practice in Mozambique. A specific set of interventions should be embraced by the project to reverse this and suggestions are made in this document on how best to go about them.

Under PROIRRI the main crops have been rice (in Zambezia and Sofala provinces), sugar cane (in Sofala) and horticulture and fruit including some cereals (mainly baby corn in Manica). The main producing modalities comprised the smallholder farmers producing for relatively open markets and self-consumption (rice, horticulture and fruit) and out growers (for sugar cane and baby corn), where smallholder farmers were subcontracted by major and well-established operators to complement their production. There were also some minor mixes among provinces, crops and production modalities.

For each of the crops PROIRRI technical assistance per crop prepared technological instructions and recommendations (Cartas Tecnológicas) outlining the main operations and the systems and procedures to be adopted to increase production and productivity.

Among other aspects (equipment, operation and maintenance of equipment and irrigation schemes, conservation, etc.) the instructions and recommendations covered the topic of fertilizers and pesticides. Lessons learned from the process are relevant to delineate how IRRIGA should tackle.

Fertilizers: the use of animal and vegetable manure (compost) was strongly encouraged as one of the ways to add organic matter to the soil to provide nutrients such as Nitrogen (N), Phosphorus (P), and Potassium (K), but in small amounts. These were adopted mainly by smallholder farmers producing for open market and self-consumption but also by outgrowers.

To obtain the maximum yield from the crops, chemical fertilizers were also recommended to be added mainly for vegetables, fruit trees and sugar cane. The most used chemical fertilizers under PROIRRI were (i) ammonium sulfate (ii) calcium ammonium nitrate (iii) super phosphate (iv) ammonium phosphate (v) urea (vi) potassium chloride (vii) potassium sulphate (viii) lime (ix) MAP and (x) KCl.

PROIRRI also encouraged the adoption of Conservation Agriculture, defined as (i) minimal disturbance of the soil (as a consequence of reduced use or no mobilization) to preserve its structure, fauna and organic matter; (ii) maintaining permanent land cover (with crops, residues and other cover materials) to protect it and contribute to the elimination of weeds; and (iii) diversification of rotations and crop combinations, which favor the maintenance of soil micro-organisms and eliminate pests, weeds and plant diseases. This was part of the integrated management of soils, water and biological resources combined with external factors

Pesticides: to begin with the project encouraged the preparation and use of homemade pesticides to lower costs while effectively undertaking the treatment of some pests. Important and recommended pesticides included: (i) petroleum emulsion, which is a contact insecticide useful against many sucking insects; (ii) tobacco mix, which is very effective for the control of aphid infesting horticultural crops; (iii) Margosa (*Azadirachta indica*) seed emulsion (neem) very effective as repellent against locusts. These were haphazardly adopted by all classes of producers and mainly the smallholder producing for open markets and self-consumption.

Additionally, the project had its own list of recommended fungicides, insecticides and herbicides to be used in crops mainly horticultures and fruit. In practical terms the most used were herbicides, namely Pendimethalin, MCPA, Metryn, Gramoxone, Glyphosate, mainly used by sugar cane growers doing outgrowing. In most cases the farmers were not directly involved in the purchase and use of such fertilizers. This was in the hands of the companies that engaged them. At times this raised conflicts between the two sides as the companies then deducted the money spent on pesticides from the final payment owed to farmers.

The environmental and social performance report (PROIRRI, 2017) identified several problems with the way fertilizers and pesticides were handled under PROIRRI. Although alternative and more affordable and healthier products including locally made products were recommended it is noticed that a consistent IPM was absent. The project was not associated with any special efforts to promote it and no systematic assessment was made. This needs to be reversed under IRRIGA.

5.3 Current pest management practices

Presently pest and plant disease control are limited by a combination of lack of knowledge, equipment, supplies and finance. In general, smallholder farmers in the project area take various measures to minimize or avoid pest infestations such as weeding and application of insecticides and herbicides. Weed control is commonly achieved through a combination of tillage-seedbed preparation by several passes of the traditional ox-drawn plough (or manually) and subsequent inter-row weed control cultivations in row crops. Comprehensive data on pesticides use are not available, but provincial officers in the subsector referred to Cypermethrin, Mancozeb, Cobox and Teodan, all under Class III (least toxic), as the main pesticides they use. These are also the pesticides mostly acquired and distributed by the public sector in critical moments.

Control of birds and wild animals (scarce in the targeted area) are mainly done by using the traditional way of scaring (the use of scarecrows is very common especially in cereal production areas), chasing and guarding of animals.

As indicated in **Tables xxxx** a combination of traditional knowledge and recommended best practices by agricultural formal agents is full of promises to the adoption of IPM.

5.4 Pesticides management

Data on pesticides poisoning and environmental contamination are often not available or difficult to obtain, since no regular government systems exist for regular monitoring of the risks. Moreover, medical personnel at rural clinics are not well trained to recognize and adequately treat pesticide poisoning, and antidotes are not systematically available in rural and in some remote provincial and district/municipal urban areas.

In summary, as related by provincial officers and the assessment report on environmental and social performance (PROIRRI, 2017), the main pesticide management problems in the project targeted area and in Mozambique in general are:

- Unfounded use when applied, which may result in problems for human health and the environment, especially the contamination of soils and water. Signs of soil/water contamination/depletion have been observed in some areas, including soil salt accumulation;
- Use of out-of-date pesticides (observed in most of areas), including by large operators who would be expected to not do so;
- Use of non-authorized and/or non-labeled pesticides or the use of re-packaged pesticides;
- Application without the adequate equipment, with an increase of the risk of contamination;
- Use of empty pesticide's packages for domestic use (e.g. as water and food containers, etc.), washed in rivers and leading to their contamination.
- Lack of adequate monitoring of pesticides use and handling is carried out.

There is a need to correct and improve current pest and pesticide management practices within the country and more in particular in the Project areas, especially given the fact that one of the objectives of the Project is to set the path to be followed in the future regarding the promotion of irrigated agriculture based on lessons learned. The plan of

action under the PMP should contribute to reverse the prevailing negative tendencies, especially those already identified under PROIRRRI.

The preliminary assessment indicates that each main area of operation in PM needs to contribute and coordinate with other areas to establish a new order that will be relevant for the project and to feed into to general PM and particularly IMP under irrigation conditions in the country.

Table 6-1: Indicative plan of action

Activity	Stakeholders	Objective/Action	Leading agency
Mobilization	INIR/DINAS, IIAM, DNEA, DPASA, SDAE Health and Environment Private sector, NGO, small, medium and large farmers	Establishment of the Core IPM Team (Representatives of Plant Diseases at the provincial and district levels, IRRIGA ESSS, Representatives of the farmers/irrigation schemes, MITADER and the MISAU) A national Workshop in the project area to identify agencies and individuals who will lead the formulation and implementation of the IPM for IRRIGA	IRRIGA assisted by DINAS
Diagnostic	INIR/DINAS, IIAM, DNEA, DPASA, SDAE Health and Environment Private sector, NGO, small, medium and large farmers	Take stock of the main issues affecting the subsector in the country and in the project area. Extensive use of lessons learned from PROIRRI Concise definition of the baseline Agree on the sequence of activities to formulate a plan of action and respective contents	IRRIGA as the convener DINAS as the leading technical department
Formulation of specific action plans	INIR/DINAS, IIAM, DNEA, DPASA, SDAE Health and Environment Private sector, NGO, small, medium and large farmers	Identification of specific issues and activities (including corrective actions vis a vis PROIRRI experiences) to be carried out in: ▪ IPM in general ▪ pesticide regulation per se, ▪ research, ▪ extension, and ▪ pesticide use	DINAS assisted by IIAM, DNEA
Implementation of action plans	INIR/DINAS, IIAM, DNEA, DPASA, SDAE	Separate and coordinated actions in: ▪ IPM in general	IRRIGA coordination

Activity	Stakeholders	Objective/Action	Leading agency
	Health and Environment Private sector, NGO, small, medium and large farmers	▪ pesticide regulation per se, ▪ research, ▪ extension, and ▪ pesticide use	In line with its role and responsibility each agency will work in its area, i.e. regulation, research, extension, IPM and pesticide use
Monitoring and evaluation and lessons learned to feed into the project and the subsector in general	INIR/DINAS, IIAM, DNEA, DPASA, SDAE Health and Environment Private sector, NGO, small, medium and large farmers	Continuous monitoring, introduction of corrective measures where needed, drawing of lessons learned and feeding back into the project and the IPM in general	IRRIGA assisted by DINAS

Table 6-3: Handling of pesticides

Nr.	Area of Action	Systems and Procedures
1	Overall	• Use resistant crops and varieties, crop sequences, associations, and cultural practices that minimize the pressure and maximize biological prevention of pests and diseases. • Maintain regular and quantitative assessment of the balance status between pest and disease and beneficial organisms of all crops. • Apply pest and disease forecasting techniques where available. • Understand and use non-chemical pest and disease management practices. • Decide on interventions following consideration of all possible methods and their short- and long-term effects on farm productivity and environmental implications to minimize the use of agrochemicals, promote integrated pest management (IPM). • Store and use agrochemicals according to legal requirements, e.g. registration for individual crops, rates, timings, and pre-harvest intervals. • Assure that agrochemicals are only applied by specially trained knowledgeable persons. • Assure that equipment used for the handling and application of agrochemicals complies with established safety and maintenance standards. • Maintain accurate records of agrochemical use. • Avoid any point source pollution from agrochemicals resulting from use, storage, cleaning and disposal of products or application equipment. • Avoid impact on non-target areas of any pest and disease management activity.
2	Procurement	• Procurement of pesticides will be the sole prerogative of licensed and authorized agents. • In line with GOM legislation in Mozambique the production, donation, trading, importation and use of any substance with a pesticide action is subject to prior registration: (i) the registration is done on the basis of a request to be submitted by companies duly registered with the DINAS (ii) the companies that register pesticides are also registered with the DINAS as pesticides importer
3	Storage	(1) pesticides shall not be stored: (a) in a manner that could result in contamination of animal feeds or commercial fertilizers. (b) above or against feed components, animal foods, foodstuffs, medications, or children's toys. (c) loaded into aerial application equipment within one hundred yards of an airline passenger terminal. (2) for display purposes: (a) pesticides shall not be shelved above or against feed components, animal foods, foodstuffs, medications, or children's toys. (b) shelving or display areas used for pesticide display shall be thoroughly cleaned before they are re-used for display of other products. (3) For liquid bulk pesticide facilities: (a) containers and accessory equipment used for the storage and handling of bulk pesticides shall be of materials and construction compatible with the pesticide stored and the conditions of storage as specified by the label instructions. (b) non-mobile liquid bulk pesticide storage container installations shall be constructed such that a secondary means of containment is provided. (i) secondary containment shall be constructed of sufficient thickness, density and composition to contain any spillage or discharged material. (ii) secondary containment areas shall not contain a drain unless it is plugged to prevent the release of any spillage from the containment area. (ii) non-mobile liquid bulk pesticide storage containers shall be secured or elevated high enough to prevent flotation if the secondary containment structure fills with liquid.

Nr.	Area of Action	Systems and Procedures
		(4) For facilities utilizing containers: (a) bulk pesticide storage and handling facilities shall be sited in accordance with appropriate local, state and national regulations and be constructed of sufficient thickness, density and composition to contain any spillage or discharged material. (b) containers and accessory equipment used for the storage and handling of bulk pesticides shall be of materials and construction compatible with the pesticide stored and the conditions of storage as specified by the label instructions. (c) containers must be protected from wind and precipitation. (d) containers must be placed on pallets or raised platforms. (e) containers must be placed on an impervious surface that extends completely beneath the pallets or raised platforms (f) the registered pesticide label shall be prominently displayed on the bulk pesticide storage container.
4	Transportation of bulk pesticide containers	(1) bulk pesticide containers shall meet all applicable standards as specified by the Decree 6/2009 of 31 March. (2) bulk pesticide containers shall be thoroughly secured to prevent spillage of pesticide and damage to containers during transportation. (3) bulk pesticide containers shall bear the registered product label for the material contained within.
5	Handling and loading of bulk pesticides	(1) bulk pesticides shall be handled, mixed and loaded in a manner that will assure the protection of crops, livestock, the public and the environment. Physical and chemical properties shall be considered in the handling and loading of bulk pesticides. (2) loading and mixing of bulk pesticides conducted at permanent distribution sites shall be on impervious surfaces with provisions sufficient to allow containment and recovery of any spillage. (a) owners or operators of permanent bulk pesticide distribution sites shall initiate immediate repair to any visible signs of damage to the impervious surfaces where bulk pesticides are loaded or mixed. (b) owners or operators of permanent bulk pesticide distribution sites shall ensure that all impervious surfaces are constructed in such a manner as to prevent water and other liquids from seeping into or flowing onto these areas from adjacent land or structures. (c) owners or operators of permanent bulk pesticide distribution sites shall ensure that all impervious surfaces for loading and mixing of bulk pesticides, are constructed with materials that are compatible with the materials being handled and are not made of asphalt or earthen materials. (3) prior to refilling, bulk pesticide containers must be thoroughly cleaned except when a sealed or dedicated recyclable bulk container is refilled with the same labeled pesticide and bears the same label as the pesticide immediately preceding it.
6	Application of pesticides	(1) the most common forms of pesticide application, in conventional agriculture, is the use of mechanical sprayers. The equipment must be in good working condition and this should be ensured by regular testing and setting of standards for application equipment
7	Disposal of containers and unused pesticides	(1) empty pesticide containers and containers which have held pesticide-treated seed: (a) shall be disposed in a manner consistent with the pesticide label; or (b) in the absence of specific label instructions, shall be triple-rinsed and disposed in designated disposal sites; and (c) shall not be sold or re-used for any purpose. Re-use of such containers for the storage of human or animal food or water, or for the storage of cooking utensils, dishes or clothing is strictly prohibited. (2) unused pesticides and unwanted pesticide-treated seeds shall be disposed in a manner consistent with their labeling.
8	Distribution of bulk pesticides	(1) bulk pesticides may be repackaged for sale or delivery if: (a) a representative of said registered establishment is present when the product is repackaged for sale or delivery; and (b) there is no change in any of the following as a result of the repackaging: (i) the pesticide formulation; (ii) the product labeling, (3) scales and meters used for bulk pesticide sales shall meet the specifications, tolerances and other technical requirements as defined by the INNOQ ³⁸ (4) separate pumps and meters shall be used for each pesticide when distributed for sale.
9	Custom mixes or blends	(1) shall be prepared to the order of the user within the recommended range of the pesticide's labeling; (2) shall not be held in inventory; (3) or portions thereof, if

³⁸ National Standardization and Quality Institute

Nr.	Area of Action	Systems and Procedures
		divided for delivery, shall be labeled with the actual percentage of the pesticide within the custom mix or blend, together with the pertinent and proper directions for its use, purpose and cautions. This requirement may be met by attaching a copy of the end-use labeling of the pesticides used in the blend. (4) shall be prepared in an establishment that is authorized by Decree 6/2009 of 31 March to deal with pesticides

Annex 5 provides a template of an IPM Plan, which should be adjusted and used at the site level to systematically assist compliance with the IPM Plan by IRRIGA subprojects.

A significant factor that can be expected to work as a constraint in the adoption of IPM practices is the attitude that pesticides are modern “medicines” that provide fast and effective cure for all problems affecting crops. Consequently, the success of any IPM strategy depends not only on the ability of the Project to define an IPM program and link it with strategic partners (private companies or NGOs), but also on the capacity of the different actors (government, extension service, farmers, private organizations, strategic partners) to fulfill their commitments in these areas. This requires considerable investment in training and capacity building in several topics of IPM and the implementation of this PMP as referred to in the main ESMF document (Chapter 11 – Training and capacity building requirements).

It is recommended that the IPM for the project will be managed and facilitated by the Hired Service Provider (HSP). The HSP will be responsible for facilitating the main stages of the process, i.e. mobilization, participatory diagnostic, formulation of the detailed plan of action involving the main actors and main areas of intervention that have been identified, implementation, monitoring and evaluation and feedback to the system through lessons learned that are valid for the project and other similar interventions. The HSP will work in close collaboration with a focal point for IPM under IRRIGA stationed at the INIR/DINAS. These will liaise with relevant agricultural operators and services including research and extension services in the fulfilment of their objectives.

Important training aspects could be done with lead farmers or involving experienced farmers. The use of lead farmers is part of the extension system in Mozambique already used in other areas of demonstration. This should be replicated accordingly under the IPM. The success of IPM will depend mostly on developing and sustaining institutional and human capacity to facilitate experiential learning for making informed decisions in integrating scientific and indigenous knowledge to minimize potential detrimental impacts of the use of pesticides. The adaptive research foreseen under the project should also include this subcomponent.

Poor communication between farmers and extension workers and other agricultural and government officers could lead to poorly-targeted research or to poor adoption of promising options generated by research. Ideally some of the training should be led by farmers themselves targeted to other farmers. Experience exchange among different farmers’ communities could prove essential for the intended outcomes.

6.2 Possible Interventions in Integrated Pest and Pesticide Management

Based on the descriptions made in previous chapters, this section provides a general outline of various types of pest control strategies known and applied in Mozambique and that can be further investigated and disseminated in wider areas, including the project area, based on evidence. These include a brief review of techniques for biological control, cultural control,

chemical control, quarantine and physical or mechanical control, chemical control and botanical control. The subchapter is complemented by three tables that summarize the best combination of traditional and modern IPM around crops and related pests as these are currently seen in Mozambique. Continuous testing and systematic drawing of lessons learned in recommended under IRRIGA.

6.2.1 STRATEGY FOR INTERVENTION AND PESTICIDE MANAGEMENT ACTION PLAN

6.2.1.1 BIOLOGICAL CONTROL

Biological control involves the use of biological agents and predators to control pests and diseases. The method is usually successful in crops like cassava and involves conservation or optimization of the impact of living agents that already exist in the ecosystem, artificially increasing the number of natural enemies in the agro-ecosystem, introducing the new natural enemies' species where these were non-existent.

Evidence shows that every living organism has its natural enemies and diseases, which keep its population at balance. Natural enemies include predators, parasitoids, nematodes, fungi, bacteria, viruses etc. The use of predators, parasitoids, nematodes, fungi, bacteria and viruses to maintain the population density of pests at a lower level than would occur in their absence is a common method under biological control or simply bio-control.

In the plant kingdom resistance to pests is the rule rather than the exception. In the coevolution of pests and hosts, plants have developed defense mechanisms. The mechanisms may be either physical (waxy surface, hairy leaves etc.) or chemical (production of secondary metabolites) in nature. Pest-resistant crop varieties either suppress pest abundance or elevate the damage tolerance level of the plant. In other words, genetic resistance alters the relationship between pest and host. The inherent genetically based resistance of a plant can protect it against pests or diseases without recourse to pesticides.

In the project area and under IRRIGA research, extension, farmers of all classes, and particularly family, small and medium size farmers will be encouraged to work together to make experiments and come up with combinations that are suitable for the area. The specific ways under which this will be done will be defined as set out in Table 6-2. Under PROIRRI and in other parts of Mozambique the use of (i) petroleum emulsion, which is a contact insecticide useful against many sucking insects; (ii) tobacco mix, which is very effective for the control of aphid infesting horticultural crops; (iii) Margosa (*Azadirachta indica*) seed emulsion (neem) very effective as repellent against locusts; (iv) intercropping with chilli pepper has proved to be effective although not always adopted systematically and not properly disseminated.

6.2.1.2 CULTURAL AND CROP SANITATION PRACTICES

Pests may also be controlled through the adoption of improved cultural and crop sanitation practices. Some of these include:

- a) **Crop rotation:** this practice is used to depress weeds and/insect pests and diseases in some crops. For example, *Striga* in sorghum and millet can be controlled/reduced by planting a trap crop like groundnuts or cotton;
- b) **Intercropping:** the field is used to grow two or more crops at the same time, which among them interchange disease control elements;

- c) **Relay cropping:** where one crop is relayed with another to reduce the infestation of weevils, for example;
- d) **Fallow:** the field is not cultivated for some years in order to control various parasitic weeds;
- e) **Cover crops:** these are leguminous crops, which are grown to suppress weeds in the field. They can be intercropped or not and they protect and cover the field e.g. pumpkins;
- f) **Trap crops:** these induce the germination of a pest. The trap crop can be intercropped or rotated with a susceptible host (e.g. groundnuts, cotton etc.).
- g) **Mulching:** this is covering of crop fields by dry grasses to control weeds and conserve soil moisture (e.g. in banana, tomato field etc.);
- h) **Hand pulling and hoes weeding:** these practices are the most common and being used by small-scale farmers. In moments of relative abundance of labor in rural areas this practice can be adopted easily;
- i) **Burning:** land clearing and destroying infected plants/crops. Although it is fundamental to ensure that burning is strictly controlled and limited to the areas and species being targeted and do not spread to other areas;
- j) **Fertilizer/manure application:** the application of nutrients in the form of either inorganic fertilizer or farm-yard manure reduces both the infestation of fields by weeds (e.g. *Striga*) and losses in crop yield;
- k) **Use of disease free planting material:** e.g. cassava cuttings, sweet potato vines etc.;
- l) **Pruning:** done in tea, orange tree etc. to reduce insect pests and diseases that might infest the crop;
- m) **Thinning:** done to reduce plant population in the field (e.g. in maize, rice, sorghum and cotton etc.).

These practices are very common and well-known in Mozambique and the project area and should be encouraged. As with biological control existing knowledge and experiments in the project area should be used and/or carried out to identify the practices that are more suitable to local conditions. Based on evidence these should be disseminated.

6.2.1.3 PHYSICAL AND MECHANICAL CONTROL

These are measures that kill the insect pest, disrupt its physiology or adversely affect the environment of the pest. They are different from cultural measures as the devices or actions adopted are directed against the insect pest instead of modifying agricultural practices. The hand picking of cotton strainers from cotton plants, banana weevils from banana pseudo stems, killing American bollworm from tomato plants are forms of physical control while the use of a fly swatter against annoying flies is a form of mechanical control as is the use of scarecrows to scare off birds, which is very common in relation to rice and wheat and other crops. Some of the mechanical measures are relatively easy to apply where and when there is abundance of manpower. This is generally the case in the targeted areas at the same time that these are well-known mechanisms.

6.2.1.4 CHEMICAL CONTROL

Chemical measures involve the use of herbicides, insecticides and fungicides to manage weeds, pests and diseases. As already explained throughout this document they should be used under certain conditions and when the other less intrusive and poisonous measures have proved to not

be effective. All the aspects of capacity building of individuals and institutions should be used to ensure that the use of chemicals is not done to the detriment of the health of humans and other living organisms and consequently ensuring a healthy environment.

They can be applied as liquid spray, in the form of granules, powder or fumigation in stores. Registered pesticides (**Error! Reference source not found.**) can be recommended as a component of IPM packages. These are registered under Pesticide Regulation (Ministerial Diploma 153/2002 of 11 September 2002) and as said updated on a yearly basis and **Error! Reference source not found.** is the list approved in December 2015 and currently in force.

6.2.2 SUMMARY OF COMBINED AND RECOMMENDED MEASURES

Pests found in agriculture and Public Health are considered all harmful organisms such as insects, weeds, or microorganisms. Current knowledge and experience combining traditional and modern IPM in Mozambique around major crops and respective pests can be summarized in the following tables. Under IRRIGA recommended practices will be adopted systematically and systematic monitoring and evaluation and drawing of lessons learned will inform progress.

Table 6-4: Recommended practices around vegetable

Pests	Name	Portuguese name	Recommendations
	<i>Aceria = Eriophyes tulipae</i>	Ácaro do Alho	- Cultural control - Biocontrol - Selective acaricide
	<i>Aculops lycopersici</i>	Ácaro do bronzeado	
	<i>Agromyza</i> sp	Minador	- Cultural control - Biocontrol - Selective insecticide
	<i>Agrotis</i> sp	Lagarta rosca	
	<i>Begrada picta</i>	Percevejo da couve	- Botanicals - Biocontrol - Resistant varieties - Selective insecticide
	<i>Bemisia spp</i>	Mosca branca	
	<i>Brevicoryne brassicae</i>	Pulgao da couve/repolho	
	<i>Diabrotica</i> sp	Crisomélidos	
	<i>Epitrix</i> sp	Crisomélidos	
	<i>Helicoverpa</i> sp	Lagarta /Broca do fruto	- Cultural control - Strip intercropping - Biocontrol - Pheromone trap - Selective insecticide
	<i>Heliothis</i> sp	Lagarta /Broca do fruto	
	<i>Lariomyza</i> sp	Lagarta mineira	
	<i>Macrosiphum euphorbiae</i>	Afideos	- Biocontrol - Selective insecticide
	<i>Myzus persicae</i>	Afideos	
	<i>Mylabris occidentalis</i>	Besouro das flores	Selective insecticide
	<i>Phyllophaga</i> spp	Escaravelho do solo	Cultural control

	<i>Plutella maculipennis</i>	Teia das couve	Biocontrol
	<i>Prodenia</i> sp	Lagartas das folhas	- Biocontrol - Selective insecticide
	<i>Solenopsis</i> sp	Formigas	- Cultural control - Selective insecticide
	<i>Spodoptera</i> sp	Lagarta dos frutos	- Biocontrol - Selective insecticide
	<i>Trichoplusia</i> sp	Lagarta medidora	
	<i>Thrips tabaci</i>	Trips	- Resistant varieties - Selective insecticide
	<i>Tetranychus</i> sp	Ácaros vermelho	Biocontrol
Diseases	<i>Aspergillus niger</i>	Podridão dos bolbos	- Phytosanitary measures - Cultural control - Resistant varieties
	<i>Alternaria brassicae</i>	Mancha zonada	
	<i>Alternaria porri</i>	Mancha púrpura	
	<i>Alternaria porri</i> f. <i>sp.dauci</i>	Queima das folhas	
	<i>Alternaria solani</i>	Mancha concêntrica/ Pinta preta	
	<i>Alternaria</i> sp	Pinta preta	
	<i>Botrytis</i> sp	Podridão da cabeça	
	<i>Bremia lactucae</i>	Míldio	
	<i>Cercospora capsici</i>	Mancha cinzenta	
	<i>Cercospora carotae</i>	Pinta cinzenta das folhas	
	<i>Cercospora</i> sp	Mancha cinzenta	
	<i>Cladosporium fulvum</i>	Mancha olivácea	
	<i>Colletotrichum capsici</i>	Antracnoses	
	<i>Colletotrichum gloesporoide</i>	Anthracnoses	
	<i>Corynespora melogenae</i>	Mancha da folha	
	Deficiência de Boro	Necroses do fruto	
	Deficiência de cálcio	Necrose ou podridão apical	
	Deficiência de magnésio	Clorose internerval	
	<i>Erwinia carotovora</i>	Podridão mole	
	Fendilhamento longitudinal da raiz	Deficiência de Boro	
	<i>Fusarium</i> sp	Murcha	
	<i>Helminthosporium</i> sp	Mancha da folha	
	<i>Leveillula taurica</i>	Míldio	
	<i>Peronospora destructor</i>	Míldio	
	<i>Phytophthora</i> , <i>Fusarium</i> , <i>Rizoctonia</i>	Damping off	
	<i>Phytophthora infestans</i>	Queima ou Míldio	
	<i>Pseudomonas solanacearum</i>	Murcha bacteriana	
	<i>Puccinia porri</i>	Ferrugem	Cultural control

	<i>Rhizoctonia solani</i>	Tombamento das mudas	• Phytosanitary measures • Resistant varieties
	<i>Rhizoctonia</i> sp	Aguado dos alfobres	
	<i>Septoria lactucae</i>	Manchas por septoria	
	<i>Septoria lycopersici</i>	Pinta da folha	
	<i>Stemphyllum botryosum</i>	Mancha amarela das folhas	
	<i>Xanthomonas axonopodis</i> pv <i>vesicatoria</i>	Queda das folhas	
	<i>Xanthomonas campestris</i>	Podridão preta/ Queima dos bordos	
Nematodes	<i>Meloidogyne</i> sp	Mal formação da raiz	• Cultural control • Phytosanitary measures • Resistant varieties
	<i>Heterodera</i> sp	Raízes bifurcadas	
Virus	Mosaic Tomato Vírus	Enrolamento do topo	• Cultural control • Phytosanitary measures • Resistant varieties
	<i>Lettuce Yellow Vírus</i>	amarelecimento borde da folhas	

Table 6-5: Recommended practices around maize, rice and sugar cane

Pests	Name	Portuguese name	Recommendations
	<i>Sesamia calamistis</i>	Brocas do colmo	• Cultural control • IPM measures • Resistant varieties
	<i>Chilo partellus</i>	Brocas do Colmo	
	<i>Busseola fusca</i>	Broca do colmo/espiga	
	<i>Rhopalosiphum maydis</i>	Afideos	• Cultural control • IPM measures • Resistant varieties
	<i>Aphis gossypii</i>	Afideos	
	<i>Agrotis segetum</i>	Roscas	
	<i>Spodoptera exempta</i>	Lagarta invasora /do colmo / maçaroca	
	<i>Acanthopplus stratiotes</i>	Matirindinde	
	<i>Zonocerus variegatus</i>	Gafanhotos	
	<i>Phymateus viripides</i>	Gafanhotos	
	<i>Heliothis</i> sp	Lagarta das espigas	
	<i>Helicoverpa</i> sp	Lagartas das espigas	
	<i>Quelea – quelea</i>	Pássaro de bico vermelho	
	<i>Mastomys natalensis</i>	Ratos	
	<i>Heterotermes</i> sp	Térmitas e Cupins	
Diseases	<i>Helminthosporium turcicum</i>	Mancha da folha	• Cultural control • Phytosanitary measures • Resistant varieties
	<i>Helminthosporium maydis</i>	Mancha da folha	
	<i>Ustilago zea</i> = <i>U. maydis</i>	Carvão comum	
	<i>Erwinia</i> sp	Podridão do colo	
	<i>Diplodia macrospora</i>	Podridão seca das espigas e grãos	
	<i>Fusarium</i> spp	Podridão das espigas	
	<i>Sphacelotheca reiliana</i>	Fungão da bandeira	

	<i>Sphacelotheca sorghi</i>	Carvão do sorgo	• Cultural control • Phytosanitary measures • Resistant varieties
	<i>Ustilago</i> sp	Carvão da espiga	
	<i>Puccinia</i> spp	Ferrugem	
	<i>Sclerospora graminicola</i>	Míldio	
	<i>Diplodia zeae</i>	Podridão do caule	
	Podridão escura da Maçaroca		
Virus	Maize Streak Vírus	Listrado da folha	

Table 6-6: Recommended practices around beans

Pests	Name	Portuguese name	Recommendations
	<i>Térmitas</i> ou Salalé	Broca da vagem	• Cultural control • IPM measures • Resistant varieties
	<i>Solenopsis</i> sp	Formigas	
	<i>Epicauta</i> sp	Besouro das folhas	
	<i>Prodenia litura</i>	Lagarta das folhas	
	<i>Myzus persicae</i>	Afideos	
	<i>Mylabris occidentalis</i>	Besouro das flores	
	<i>Omphyomia phaseoli</i>	Minador/ Fendilhamento do caule	
	<i>Agrotis</i> sp	Roscas	
	<i>Acanthoscelides obtectus</i>	Lagarta das vagens	
	<i>Prodenia litura</i>	Lagarta das folhas	
	<i>Aphis fabae/ Aphis cracivora</i>	Pulgão preto do feijoeiro	
	<i>Ootheca mutabilis</i>	Crisomélido das folhas	
	<i>Epicauta velata</i>	Besouro da folhas	
	<i>Bemisia tabaci</i>	Mosca branca	
	<i>Tertranychus</i> sp	Ácaros vermelhos	
	<i>Spodoptera</i> spp	Lagarta das vagens	
	<i>Heliothis</i> spp	Lagarta das vagens	
	<i>Anoplocnemis</i> sp	Percevejo das folhas	
Diseases	<i>Cercospora arachidicola</i>	Mancha castanha	• Cultural control • Phytosanitary measures • Resistant varieties
	<i>Cercospora personata</i>	Mancha preta arredondadas	
	<i>Puccinia arachidis</i>	Ferrugem	
	<i>Alternaria arachidis</i>	Mancha zonada	
	<i>Rhizoctonia</i> sp	Podridão seca da raiz	
	<i>Ascochita</i> sp	Mancha por Ascochita	
	<i>Colletotrichum</i> sp	Anthraxnose das folhas	
	<i>Pseudomonas</i> sp	Mancha bacteriana	
	<i>Uromyces appendiculatus</i>	Ferrugem	
	<i>Colletotrichum lindemuthianum</i>	Anthraxnose	• Cultural control • Phytosanitary measures • Resistant varieties
	<i>Glomerela cingulata</i>	Anthraxnose	
	<i>Alternaria tenuissima</i>	Mancha concêntrica das folhas	
	<i>Phoma</i> sp	Mancha de Ascoshyta	
	<i>Phaeoisariopsis griseola</i>	Mancha angular	
	<i>Erysiphe polygoni</i>	Míldio	
	<i>Meloidogyne</i> spp	Nematodos de lesao ou de galha	
Nematodes			
Virus		Mosaico comum do Feijoeiro -BCMNV	• Cultural control • Phytosanitary measures • Resistant varieties

6.3. AUTHORIZED PESTICIDES

If the project will not completely switch to and enforce a full organic approach, it is inevitable that pesticides will be recommended for use on some sites and crops. In the territory defined by the Project area the use of agro-chemicals, including pesticides is already a reality and growing. There are pockets of farmers that doing so and if no other measures are taken this will tend to grow.

Based on what happened under PROIRRI, when compared with other areas in Mozambique the project area can rank high at the national level in the use of agro-chemicals. This is mainly associated with the production of sugar cane, baby corn, fruits and horticulture. With the planned support to agricultural intensification the use of agro-chemicals can be expected to be intensified.

Under the World Bank funding for the Project, no funding for pesticide acquisition will be provided for farmers. However, it would be recommended to elaborate a provisional list of less harmful pesticides that can be used. A list of registered pesticides in Mozambique is provided as an Annex of the Pesticide Regulation (Ministerial Diploma 153/2002 of 11 September 2002) and includes among others: cypermethrin, deltamethrin, mancozeb and dimethoate. The list is updated on a regular basis and **Error! Reference source not found.** of this document presents the latest version updated in December 2015. Annex 4 of this document presents the WHO Pesticide Classification List by level of hazardousness. The list could guide the classification of pesticides eventually to be used. Annex 5 outlines the site-specific plan to implement IPM at the subproject level. It is provided as a sample plan to be adjusted and used in different subproject contexts.

7. INSTITUTIONAL STRENGTHENING, TRAINING AND CAPACITY BUILDING

Different measures will have to be combined to mitigate the potential adverse impacts likely to occur because of pesticide use in the project areas. Ideally the Department under the DINAS dealing with plant diseases should be better represented in IRRIGA provinces and districts in a way that goes beyond what is currently the practice. There are less than 10 technical staff members at the central level, 2-3 at the provincial level and at the district level the unit is represented by extension workers or any other officials who cumulatively develop other activities. Additionally, the units are ill-equipped in terms of other means to develop their work (transport, laboratories, and other relevant equipment). Support will be provided by the project to have clearly designated officers at both provincial and district level to provide dedicated work to IPM.

IRRIGA IPM will be managed in line with the institutional arrangement of the GOM, this project and the MASA, as described and assessed in Chapter 4. The process involves the relevant government departments (MASA, MISAU, MITADER, and MIC) at central, provincial and district levels. Within the MASA it will actively involve DINAS and its Department of Plant Diseases, the Research Institute (IIAM) and Extension Services. Other national research and higher education institutions (e.g. UEM) will also be involved. The project will take advantage of the fact that the unit dealing with plant diseases is within the same ministry as INIR.

On a day to day basis PIU at Central and Provincial level will rely on their agronomists and the Environmental and Social Safeguards Specialists and all other MASA agents at provincial and district levels to take responsibility for all IPM requirements. This will also extend to TA around the crops to be embraced by the project as was the case under PROIRRI. These will be assigned specific responsibilities for IPM development and implementation. At the subproject level the Project Managers, Agricultural Technical Officers and/or other designated people will have direct responsibility for IPM. Annex 5 will be adjusted to guide their day-to-day work.

The primary mitigation measure includes institutional strengthening and the second is the training of the various categories of stakeholders in the pesticide chain in safe and thoughtful pesticide use and management. The latter includes the delivery of a mix of (i) Information, Education and Communication approaches targeting farmers, pesticide operators and teams; (ii) provision of Personal Protective Equipment (PPE); (iii) training to farmers, and thorough and consistent supervision and monitoring. Specific aspects and numbers of people to be involved will be defined after the diagnostic of the situation and preparation of the final version of the plan of action foreseen in Table 6-1.

7.1 Institutional Strengthening

IPM success relies significantly on developing and sustaining institutional and human capacity to facilitate experiential learning for making informed decisions in integrating scientific and indigenous knowledge to solve specific problems. In this regard IRRIGA is relevant by adopting adaptive research.

Poor communication between farmers, extension workers has often led to poorly-targeted research or to poor adoption of promising options generated or that could be generated by research.

A sound IPM in Mozambique can only be achieved through a good and practical combination of scientific and applied/participatory research involving farmers (including women), extension workers, and researchers. Under IRRIGA stakeholders will need to get together and in a process with multiple stages agree on issues to be addressed and define a plan of action, implement and monitor it. During the formulation of this plan it was not possible to bring all of them together and the information obtained from each of the few that were directly contacted cannot substitute this process of formulation of an agreed plan of action. The issues to be addressed are complex and require a good combination of research, plan and action, lessons learned and so on.

INIR/IRRIGA in coordination with DINAS and the latter through its Plant and Animal Health Services, are required to work together with the IIAM (research stations in the project area (e.g. Chimoio and Nampula and other)) and other related sectors (health and environment) to institute research programs that respond to local needs. For the programs to be adequately responsive they should involve local farmers (including women) and extension workers in the identification of issues to be worked on and subsequent delivery of responses and ways of putting them into practice, including the training, capacity building and skills and attitude development. The “on farm” and “on station” research programs embraced by IIAM should be creatively used to work for this objective.

The private sector and local relevant NGOs should also be involved. The program should also include initiatives such as Farmer Field Schools, Training of Trainers, and regional meetings etc., which could bring together actors from different projects (e.g. SUSTENTA) but with similar objectives in this area.

PROIRRI used the above-described structure to build capacity on irrigation in general and now it should create a line of work to specifically address PM and irrigation.

7.2 Training and Capacity Building

This is not a straight forward perception but in general farmers are aware that pesticides are poisonous and represent a serious occupational health and environmental risk. Also, the cost of pesticides is discouraging for most poor farmers to embark on massive use. To take advantage of this conducive situation there is a need to implement awareness campaigns to raise the understanding of the potential environmental and human health impacts related with inadequate use of pesticide as well as the benefits of the integrated pest management practices.

Capacity building will be achieved through farmer-based collaborative management mechanisms under which all key stakeholders have to be regarded as equal partners whose role will be to facilitate the process and provide technical direction and any other support necessary for the implementation of the activities

The HSP backed by INIR/IRRIGA and DINAS Focal Point will take the necessary steps to prepare comprehensive training manuals, brochures and leaflets on pesticide use and management, targeting different actors within the project, ranging from extension service providers, farmers, loaders, mixers, transporters, government staff among others. The training manual or guides to be developed for use must be simplified and easy to understand and participatory in nature with in-built and demonstration/ practical sessions as much as possible.

More specifically training on IPM, targeting lead farmers, extension workers, local leaders, etc. will include but not be limited to:

- **Learning-by-doing/discovery training programs:** farmers are most apt to adopt new techniques when they acquire knowledge and skills through personal experience, observation, analysis, experimentation, decision-making and practice. This allows to identify farmers' own knowledge and for farmers to understand how IPM applies to their own farms.
- **Recovering collective memory:** pest problems often emerge because traditional agricultural methods were changed in one way or another or lost. These changes can sometimes be reversed. This approach uses group discussions to try to identify what changes might have prompted the current pest problem.
- **Focus groups discussions:** regular meetings among women, men, the youth to discuss production problems including pests and related problems can assist in the success of various control methods. These meetings should be promoted using all forms of local incentives.
- **Demonstration projects:** farmer-field schools can be very effective at promoting IPM within the local community. These pilot sessions demonstrate IPM in action and allow farmers to compare IPM with ongoing cultivation supported by synthetic pesticides.
- **Educational material:** basic written and photographic/figures guides or even videos about pest identification and crop-specific management techniques are essential for training and could be an important factor in motivating farmers to adopt IPM.
- **Youth education:** promoting and improving the quality of programs on IPM and the risks of synthetic pesticides has been effective at technical schools for rural youth. In addition to becoming better farmers in the future, these students can bring informed views back to their communities.

Training in the use of pesticides, targeting lead farmers, extension workers, local leaders, etc., will include but not be limited to:

- **Pesticide selection:** indicating the list of authorized pesticides per target pests, indicating their level of toxicity and hazardous, possible harmful effects and experience of using those pesticides for the pest and the crop.
- **Understanding the pesticide label:** explaining all the information included in the label. Pesticide transport: give indications on how to transport pesticides to avoid any leakages and avoiding the contact with persons or animals.
- **Mixing and loading pesticides:** explain the importance of ensuring the proper dilution of the concentrated pesticide and the need to use protective clothing.
- **Pesticide storage:** give indications on how to storage pesticides (i.e. site location (not allowed in flood areas), security (against illegal entries, as well as children and livestock), isolated from housing, well ventilated, waterproof roof, have a current inventory list of pesticide stock.
- **Container disposal:** giving indications on how to destroy/dispose used pesticide containers
- **Obsolete pesticides:** explain the risks associated with obsolete pesticides and procedures to be followed.
- **Calibration, product quantity and pesticide application:** explaining the importance of application equipment calibration and how to do it.
- **Determining the amount of chemicals to use:** giving explanations on methods to find out the amount of chemicals to apply per hectare and levels of dilution
- **Precautions related to the application of pesticides:** giving indications on important precautions for safe use of pesticide

- **Toxicity, human protection and first aid:** explaining the possible effects of pesticide on human health, ways of pesticides entering in the body, importance of protective clothing and other protective equipment, basic first aid for pesticide exposure (with skin, mouth, eye or respiratory system).

Under IRRIGA the distribution and use of pesticides to or by people who have not received training must be prevented. Inspection and monitoring work foreseen under the ESMF should also include a line of work to take care of this aspect.

All the details on institutional strengthening and training and capacity building should be revisited at the start of the project as suggested in Chapter 6.